SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I G.GOWTHAMI(192524285) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.GOWTHAMI

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ANSWERS

Task 1: Article Summary

(a) Article Citation, Domain and ML Problem

Sample Article Citation:

Kavakiotis, I., Tsave, O., Salifoglou, A., Maglaveras, K., Vlahavas, I., & Chouvarda, I. (2017). Machine Learning and Data Mining Methods in Diabetes Research. Computational and Structural Biotechnology Journal, 15, 104–116. DOI: 10.1016/j.csbj.2016.12.005.

Domain/Application Area:

The article belongs to the healthcare and medical informatics domain, specifically diabetes diagnosis, prediction, monitoring, and treatment support.

Specific ML Problem:

The main problem is to apply Machine Learning (ML) and Data Mining techniques to medical data to identify patterns associated with diabetes and support better diagnosis, prediction, and management of the disease.

The article discusses how different machine learning approaches can analyse large volumes of healthcare data and assist healthcare professionals in making informed decisions. It focuses on techniques such as classification, clustering, regression, decision-tree-based approaches, neural networks, and other data-mining techniques.

(b) Key Objective, Research Gap and Proposed Solution

The main objective of the article is to examine how machine learning and data-mining methods can be applied to diabetes-related healthcare problems. Diabetes produces large amounts of clinical and patient-related data, including blood glucose measurements, demographic information, medical history, laboratory results, and treatment information. Traditional approaches may not always be able to identify complex relationships hidden within these datasets. The authors therefore investigate how computational learning methods can extract useful patterns from diabetes data and support diagnosis, prediction, monitoring, and treatment. A major research gap identified is the lack of a unified understanding of how different machine learning and data-mining techniques are being applied across different diabetes-related tasks. The article addresses this gap by reviewing and analysing existing computational approaches and their applications in diabetes research. It highlights the advantages and limitations of different algorithms and demonstrates that machine learning can support personalised healthcare by identifying high-risk patients and predicting disease-related outcomes. The study also emphasises the importance of appropriate datasets, feature selection, validation, and clinical interpretation when applying ML to healthcare.

Task 2: Methodology Analysis

(a) ML Techniques Used

The article is primarily a review and analysis of machine-learning and data-mining approaches rather than a single experimental implementation of one algorithm.

Several important ML approaches relevant to diabetes research are discussed.

1. Decision Trees

Decision Trees classify patients by repeatedly splitting the dataset according to important features.

For example:

```
          Glucose
         /     \
      Low        High
     /             \
  Normal          BMI
                 /  \
              Low    High
               |      |
           Non-Diabetic Diabetic
```

Decision Trees are useful in healthcare because their decisions can be represented as understandable rules.

2. Statistical Learning

Statistical methods such as Logistic Regression can estimate the probability that a patient belongs to a particular class.

For diabetes prediction:

where:

Features may include glucose, BMI, age, blood pressure and family history.

3. Classification

Classification algorithms attempt to assign patients to categories such as:

Diabetic

Non-diabetic

High risk

Low risk

Different algorithms can be compared according to accuracy, sensitivity, specificity and other metrics.

4. Clustering

Clustering is an unsupervised learning approach. Patients can be grouped according to similarities in their health characteristics without requiring a predefined class label.

For example:

Patient Data

↓

Clustering

↓

Cluster 1 → Low-risk patients

Cluster 2 → Moderate-risk patients

Cluster 3 → High-risk patients

Dataset

Diabetes research can use datasets containing patient demographic, clinical and laboratory information.

Typical features include:

Feature

Description

Age

Patient age

Glucose

Blood glucose concentration

BMI

Body Mass Index

Blood Pressure

Blood pressure measurement

Insulin

Insulin level

Pregnancy

Number of pregnancies

Skin Thickness

Skin-fold measurement

Diabetes Pedigree

Family/genetic risk indicator

Outcome

Diabetes diagnosis

The data generally requires preprocessing before ML modelling.

Data Preparation

Typical preprocessing includes:

Removing duplicate records.

Handling missing values.

Detecting abnormal values.

Encoding categorical variables.

Normalising numerical features where required.

Selecting relevant features.

Splitting data into training and testing sets.

Handling class imbalance.

(b) Mapping to CO4 Topics

CO4 Topic

Connection with Article

Inductive Learning

Models learn general patterns from historical patient records and use them to predict new cases.

Decision Trees

Tree-based rules can classify patients according to clinical features.

Statistical Learning

Logistic Regression and statistical methods can estimate disease probability and risk.

Reinforcement Learning

Sequential patient-treatment decisions can potentially be modelled using states, actions and rewards.

Methodological Strength

One major strength is the wide coverage of machine-learning methods and their applications to diabetes research. Comparing different approaches helps researchers understand which methods may be suitable for particular healthcare problems.

Another strength is the emphasis on using data-driven approaches to identify hidden relationships between patient characteristics and diabetes outcomes.

Methodological Limitation

A major limitation is that machine-learning performance can vary significantly depending on the dataset, preprocessing method, feature selection, and evaluation strategy.

A model achieving high accuracy on one dataset may not necessarily perform equally well on patients from another hospital, geographical region, or demographic population.

Therefore, external validation is extremely important before clinical deployment.

Task 3: Results & Critical Evaluation

(a) Key Results

Because the article is primarily a review, it discusses results from multiple studies rather than reporting one single universal accuracy.

Different ML studies have reported different performance levels depending on the dataset and algorithm.

An illustrative comparison is:

Model

Accuracy

Precision

Recall

F1-Score

AUC

Decision Tree

85%

83%

84%

83.5%

0.88

Logistic Regression

87%

85%

86%

85.5%

0.90

Random Forest

89%

88%

87%

87.5%

0.92

Neural Network

91%

89%

90%

89.5%

0.94

Note: These numbers are an illustrative table for assignment explanation and should not be presented as exact results from the article unless they appear in the paper.

Interpretation

The table demonstrates that different algorithms can produce different results.

Accuracy measures overall correct predictions.

Precision measures how many predicted diabetic cases were actually diabetic.

Recall measures how many actual diabetic cases were successfully detected.

F1-score balances precision and recall.

AUC-ROC measures the ability to distinguish between diabetic and non-diabetic patients across thresholds.

For medical diagnosis, Recall is particularly important because false negatives may cause diabetic patients to be missed.

(b) Critical Evaluation

Are the Results Realistic?

High accuracy values in diabetes prediction are possible, but they should be interpreted carefully.

For example, an accuracy of 95% may initially appear excellent. However, if 90% of the dataset is non-diabetic, a model that predicts almost everyone as non-diabetic could achieve high accuracy while failing to detect diabetic patients.

Therefore, accuracy should not be considered alone.

The study should ideally report:

Accuracy

Precision

Recall/Sensitivity

Specificity

F1-score

AUC-ROC

Confusion matrix
Calibration
External validation results
Potential Dataset Bias
Healthcare datasets can contain several types of bias.
1. Sampling Bias
If the dataset is collected from only one hospital, it may not represent the entire population.
2. Demographic Bias
Age, gender, ethnicity, socioeconomic conditions and geographical factors may affect disease patterns.
3. Class Imbalance
If diabetic cases are much fewer than non-diabetic cases, the model may favour the majority class.
4. Measurement Bias
Different hospitals may use different equipment or laboratory procedures.
5. Historical Bias
Historical medical records may contain existing inequalities or differences in healthcare access.
Evaluation Issues
A stronger evaluation would include:
Stratified K-fold cross-validation
External validation dataset
Independent hospital dataset
Confusion matrix analysis
Sensitivity and specificity analysis
Calibration testing
Subgroup performance analysis
Comparison with existing clinical risk scores
These experiments would demonstrate whether the model generalises beyond the original dataset.
(c) Real-World Applicability
Machine learning for diabetes can be applied in several areas.
1. Hospital Decision Support
The model can assist doctors in identifying patients who may require additional diabetes testing.
2. Early Screening
High-risk individuals can be identified before serious complications develop.
3. Personalised Healthcare
Patient characteristics can be used to generate personalised risk assessments.
4. Remote Monitoring
Wearable devices and home glucose-monitoring systems can continuously provide health information.
5. Electronic Health Records
ML models can analyse historical patient records and automatically identify high-risk cases.
Industry Challenges
Data Availability
Large, high-quality and properly labelled medical datasets are difficult to obtain.
Privacy
Patient information must be protected against unauthorised access.
Scalability
Large healthcare organisations may need significant storage and computing resources.
Model Generalisation

A model trained using one hospital's data may not work equally well in another hospital.

Explainability

Doctors need to understand why an ML model predicts that a patient is at high risk.

Ethical Issues

The system must avoid discrimination and should not make unsafe treatment decisions without medical supervision.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1: Inductive Learning

The first important insight is that machine learning performs inductive reasoning by learning general patterns from specific examples.

For example, the model can learn from thousands of historical patient records and then estimate diabetes risk for a new patient.

CO4 Connection:

This directly relates to Inductive Learning, where a general hypothesis is learned from observed examples.

Insight 2: Decision Trees Provide Explainability

The second insight is that Decision Trees are useful in healthcare because their predictions can be represented as understandable rules.

For example:

IF glucose is high

AND BMI is high

AND age is above a certain level

THEN diabetes risk is high.

This is easier for clinicians to understand than an entirely black-box prediction.

CO4 Connection:

This relates directly to Decision Tree learning, splitting criteria, information gain and classification.

Insight 3: Accuracy Alone Is Not Enough

The third insight is that a medical ML model cannot be evaluated only using accuracy.

A model may have high accuracy while missing many diabetic patients. Therefore, Recall, Precision, F1-score, AUC-ROC and confusion-matrix analysis are important.

CO4 Connection:

This relates to Statistical Learning and model evaluation, where different performance measures are used to understand classification quality.

(b) Proposed Novel Extension

Proposed Experiment: Hybrid ML + Reinforcement Learning for Personalised Diabetes Management

A useful extension would be to combine a diabetes risk-prediction model with Reinforcement Learning.

The first model could predict the patient's current risk:

Patient Data

↓

ML Risk Prediction

↓

Low / Medium / High Risk

↓

RL Agent

↓

Diet / Exercise / Monitor / Clinical Review

↓

Patient State Change

↓

Reward

↓

Updated Policy

Proposed Dataset

A larger longitudinal dataset containing repeated measurements could be used instead of only one-time patient records.

Possible information includes:

Blood glucose

BMI

Blood pressure

Physical activity

Diet information

Medication history

Age

Family history

Previous health state

Follow-up measurements

Proposed Algorithm

A Deep Q-Network (DQN) could be used when the patient state becomes too large for a simple Q-table.

The state could contain:

Actions could include:

The reward could reflect improvement in validated health outcomes while strongly penalising unsafe outcomes.

Expected Impact

The proposed system could move beyond simply answering:

“Does this patient have a high diabetes risk?”

and investigate:

“Given the patient's changing health state, what safe management strategy should be considered next?”

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand